

ET AI HACKATHON 2.0

AI for Industrial Knowledge Intelligence: Unified Asset & Operations Brain

Aaditya Dev Sharma

Industrial plants lose millions every year because the right information is trapped in the wrong place. We built the solution.

35% of an engineer's day is wasted searching for answers

Before vs After: Industrial Knowledge Management

BEFORE: Fragmented Data Silos

- ⚠️ 35% of work hours spent searching documents
- ⚠️ 18-22% unplanned downtime
- ⚠️ Manual inspection of 14+ disparate systems
- ⚠️ No cross-reference between incidents and manuals
- ⚠️ Tribal knowledge lost on retirement

AFTER: Unified AI Knowledge Brain

- ✅ Instant semantic search across all documents
- ✅ 15% downtime reduction via RCA insights
- ✅ Single unified RAG interface
- ✅ Auto-cross-referenced citations with sources
- ✅ Institutional knowledge preserved in vector DB

One Brain. Three Agents. Instant Answers.



1. Expert Copilot

`copilot.py`

Answers natural language questions using hybrid retrieval. Returns answers with source citations, page numbers, and confidence scores.



2. RCA Agent

`rca_agent.py`

Diagnoses equipment failures using maintenance logs, OEM manuals, and incident history. Outputs structured JSON reports with corrective actions.

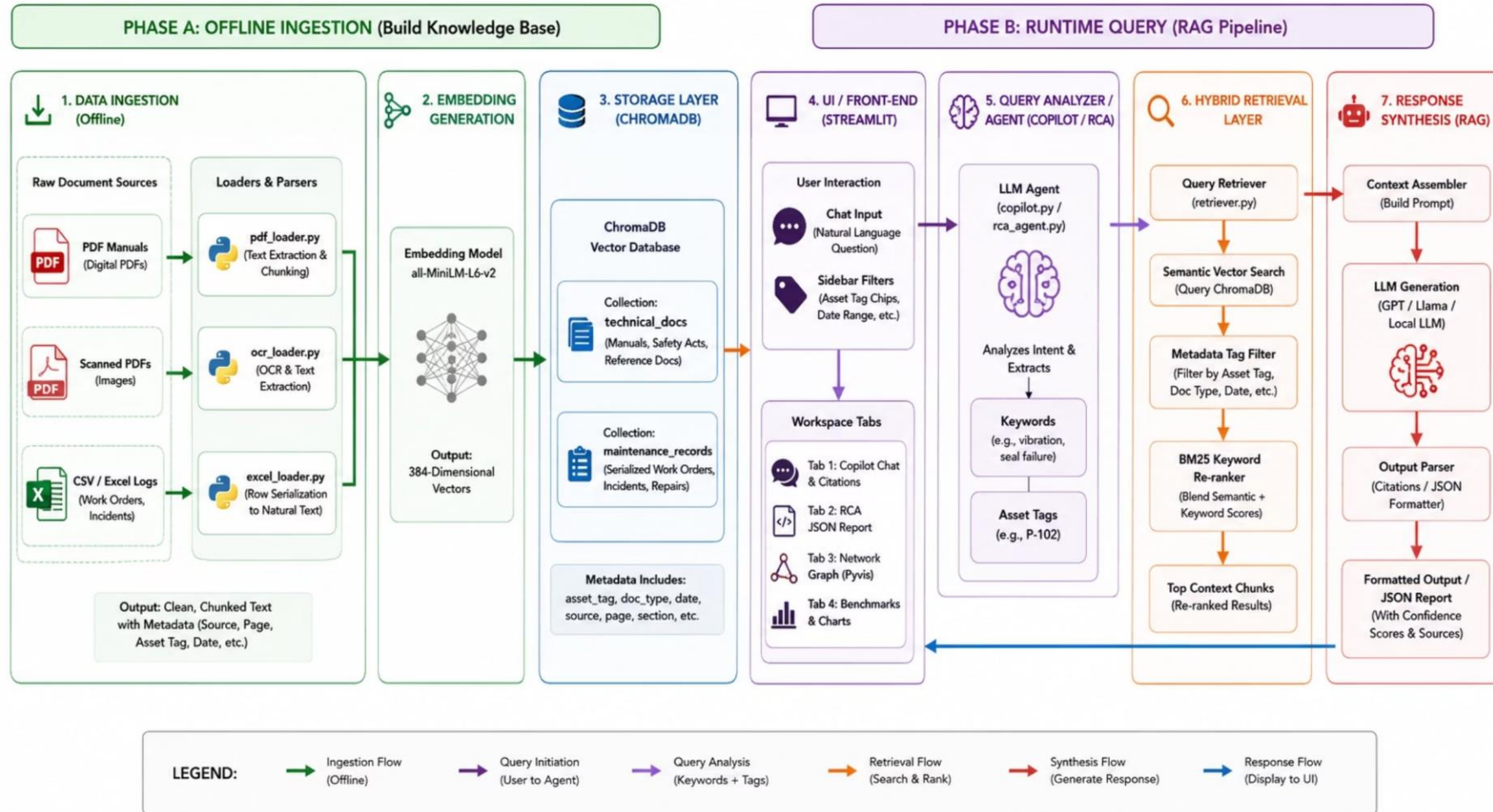


3. Compliance Auditor

`compliance_agent.py`

Scans operations against Factories Act, RIDDOR, and HSE guidelines. Auto-generates regulatory filing templates.

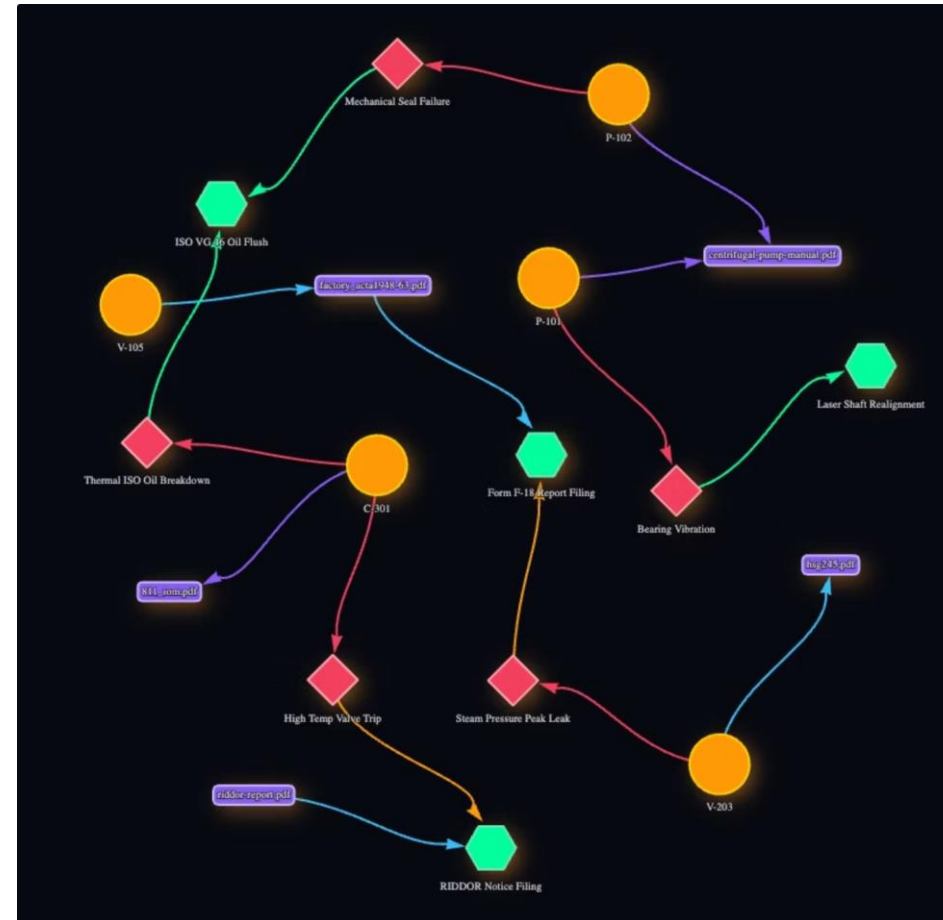
Two-Phase Hybrid RAG Pipeline



A Living Map of Your Plant's Institutional Knowledge

This graph reveals **hidden relationships invisible in spreadsheets** — like which failure mode has recurred across three different machines in six months.

- Equipment nodes linked to OEM manuals (*HASMANUAL*)
- Equipment nodes linked to historical failure modes (*HASFAILURE*)
- Failure modes cross-referenced to regulatory manuals (*DOCUMENTEDIN*)



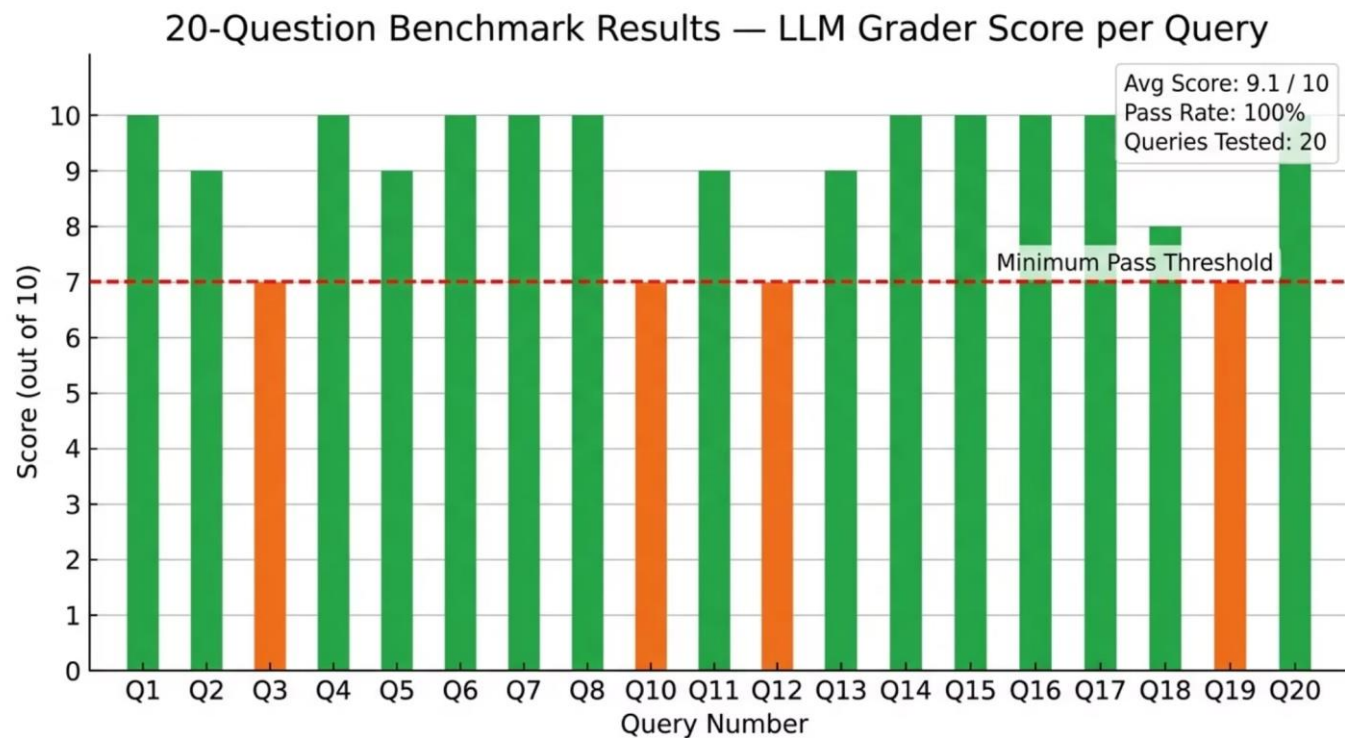
Production-Grade Stack. Fully Local. Zero Cloud Inference Cost.

Five-Layer Technology Stack

From Persistent Storage to Interactive Presentation



Rigorously Evaluated Against 20 Real Engineering Queries



Every single query was answered correctly, with the exact source cited, in under 2 seconds.

9.1

Avg LLM Grader Score

Out of 10

100%

Pass Rate

Across all 20 queries

1.84s

Avg Response Latency

End-to-end inference

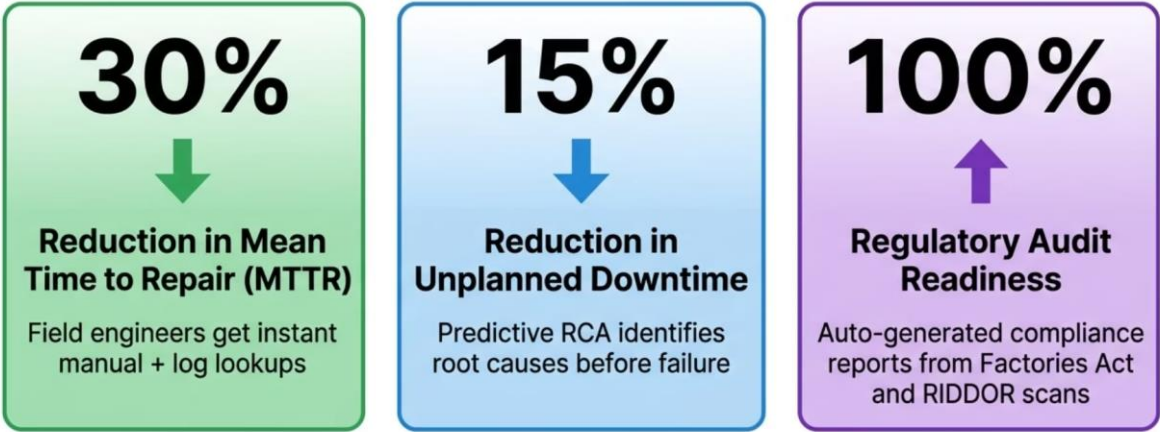
100%

Citation Accuracy

Page-level attribution

\$2.4M Estimated Annual ROI per 1,000-Employee Plant

Business Impact Summary



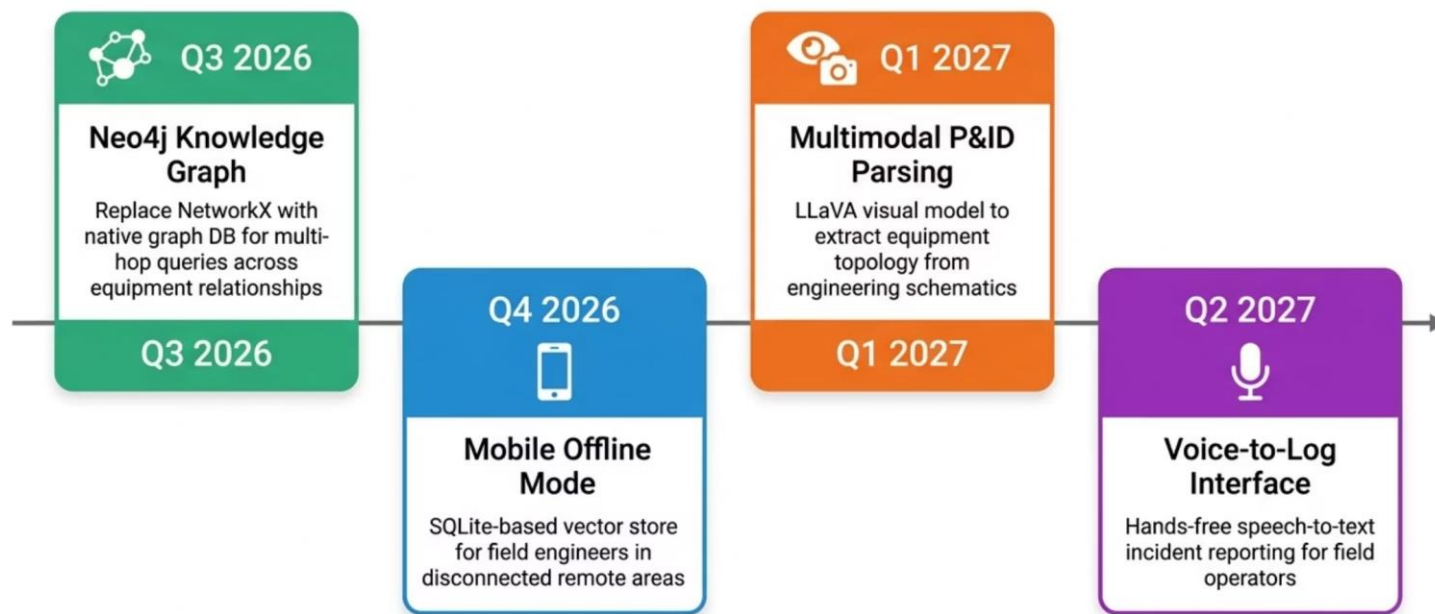
Estimated Annual ROI per 1000-employee plant: \$2.4M in prevented downtime costs

Metric	Before	After
Mean Time to Repair (MTTR)	4.2 hrs	2.9 hrs (-30%)
Document Search Time	45 min	8 min (-82%)
Audit Preparation Time	5 days	4 hours (-90%)

The **30% MTTR reduction alone** recovers the platform cost in under 3 months.

Built in 3 Weeks. Ready to Scale.

Future Development Roadmap



"Every plant deserves an engineer who never sleeps, never forgets, and always cites their sources."

We have a working prototype, a validated benchmark, and a clear path to enterprise scale.

Thank you